SURJECTIVITY OF THE GLOBALLY GENERATED SUBMODULE ONTO A ZERO-DIMENSIONAL QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the closing contradiction 'H^0(X, F) is bigger than H^0(X, F')' (source lines 1399-1401) is resolved by the definition of F' as the submodule generated by ALL global sections of F: any global section of F factors through F', so Gamma(F) = Gamma(F') and the quotient H^0(F)/H^0(F') is zero; no external input is needed for this step.

Editorial note. The label lemma-globally-generated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cohomological vanishing for the constructed submodule. Same chapter, \label{lemma-vanishing}. Proof lines 1391-1393: H^1(X, F') = 0 follows from this lemma and the assumption that no nonzero submodule of F injects into Q.
- The Cohen-Macaulay curve situation. Same chapter, Situation \label{situation-Cohen-Macaulay-curve}. Statement line 1372: the lemma is set in this situation; in particular dim_k H^1(X, omega_X) = dim_k H^0(X, O_X) = 1 (used in proof lines 1387-1389).

Source reference. Original reference 07LA: sources/stacks-curves/pinned-original-source.tex, source lines 1321–1321. Prerequisite pointer invoked at proof line(s) 1392; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B5D: sources/stacks-curves/pinned-original-source.tex, source lines 1307–1307. Prerequisite pointer invoked at proof line(s) 1373; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0B5D. Given an exact sequence*

$$\omega_X \to \mathcal{F} \to \mathcal{Q} \to 0$$

of coherent $\mathcal{O}_X$-modules with $\dim(Supp(\mathcal{Q})) = 0$ and $\dim_k H^0(X, \mathcal{Q}) \geq 2$ and such that there is no nonzero submodule $\mathcal{Q}' \subset \mathcal{F}$ such that $\mathcal{Q}' \to \mathcal{Q}$ is injective. Then the submodule of $\mathcal{F}$ generated by global sections surjects onto $\mathcal{Q}$.

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Proof. Let $\mathcal{F}' \subset \mathcal{F}$ be the submodule generated by global sections and the image of $\omega_X \to \mathcal{F}$. Since $\dim_k H^0(X, \mathcal{Q}) \geq 2$ and $\dim_k H^1(X, \omega_X) = \dim_k H^0(X, \mathcal{O}_X) = 1$, we see that $\mathcal{F}' \to \mathcal{Q}$ is not zero and $\omega_X \to \mathcal{F}'$ is not an isomorphism. Hence $H^1(X, \mathcal{F}') = 0$ by Lemma 07LA and our assumption on $\mathcal{F}$. Consider the short exact sequence

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{Q}/\operatorname{Im}(\mathcal{F}' \to \mathcal{Q}) \to 0$$

If the quotient on the right is nonzero, then we obtain a contradiction because then $H^0(X, \mathcal{F})$ is bigger than $H^0(X, \mathcal{F}')$. $\square$